LLM Legal Complaint Analyzer

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Introduction

Legal professionals spend significant time on early-stage case assessment, including reviewing complaints, identifying key legal issues, and formulating initial discovery questions. Legal complaints are long, dense, and often time-sensitive, and can require up to 3–6 hours of review per case, depending on the case's complexity. The time-intensive nature of this review process creates a bottleneck in case assessment and can delay the development of an initial legal strategy.

The **LLM Legal Complaint Analyzer** tool aims to assist with this process by transforming a multi-hour manual workflow into an automated pipeline that produces concise summaries, structured insights, and actionable attorney next steps within seconds. The tool enables a user to upload a legal document of their choosing and receive two main outputs in return:

1) A structured summary of the case
2) A set of actionable next steps designed for an attorney

The tool is designed as a triage assistant rather than a replacement for legal judgment. Its value is not that it makes final legal decisions, but that it organizes the document quickly enough for a legal professional to understand what deserves closer review and follow-up.

The current prototype focuses on initial legal complaints because they are a common starting point for litigation review. Complaints usually contain enough information to identify the basic dispute, parties, claims, requested relief, and early factual theory. At the same time, they can be difficult to skim because relevant facts are spread across jurisdictional allegations, factual background sections, numbered counts, and exhibits or attachments. This made complaints a useful document type for testing whether an LLM could separate legally important information from less useful background material.

Methodology / Architecture

The LLM Legal Complaint Analyzer is a full-stack application built in React (front end) and Python (back end). The front end allows document upload and displays the summary and attorney next steps in a structured layout. The back-end handles file extraction, prompt construction, model calls, and response formatting.

The tool workflow begins when the user selects a legal document from their local folders to upload. The front end accepts common legal document formats such as PDF, DOCX, and TXT. The selected file is then sent to the backend for analysis. Front end development in particular was assisted by AI tools (after receiving permission from Dr. Madiraju), as I wanted to focus my learning on LLM capabilities and less so on front end development and website design.

The Python + FastAPI backend coordinates the document processing workflow and LLM generation. When a file is received, the document processing pipeline reads the file content and extracts plain text used as the input for the LLM. Text-based PDFs are processed with pypdf, DOCX files are processed with python-docx, and TXT files are decoded directly. After extraction, the backend sends the resulting text into the LLM pipeline for summarization.

The LLM selected for this project is **LLaMA 3.1 8B**. Because LLaMA models require permissioned access, access to the model had to be requested before it could be used in the application. After access was approved, the application could then connect to the model using a Hugging Face API token. Configuration values such as the Hugging Face token, model ID, and chat completions endpoint are loaded from environment variables. If an API token is unavailable, the backend returns mock summary and next-step text. This fallback is useful for development because the application can still be demonstrated without making live model calls, but the real intended behavior still depends on the Hugging Face API.

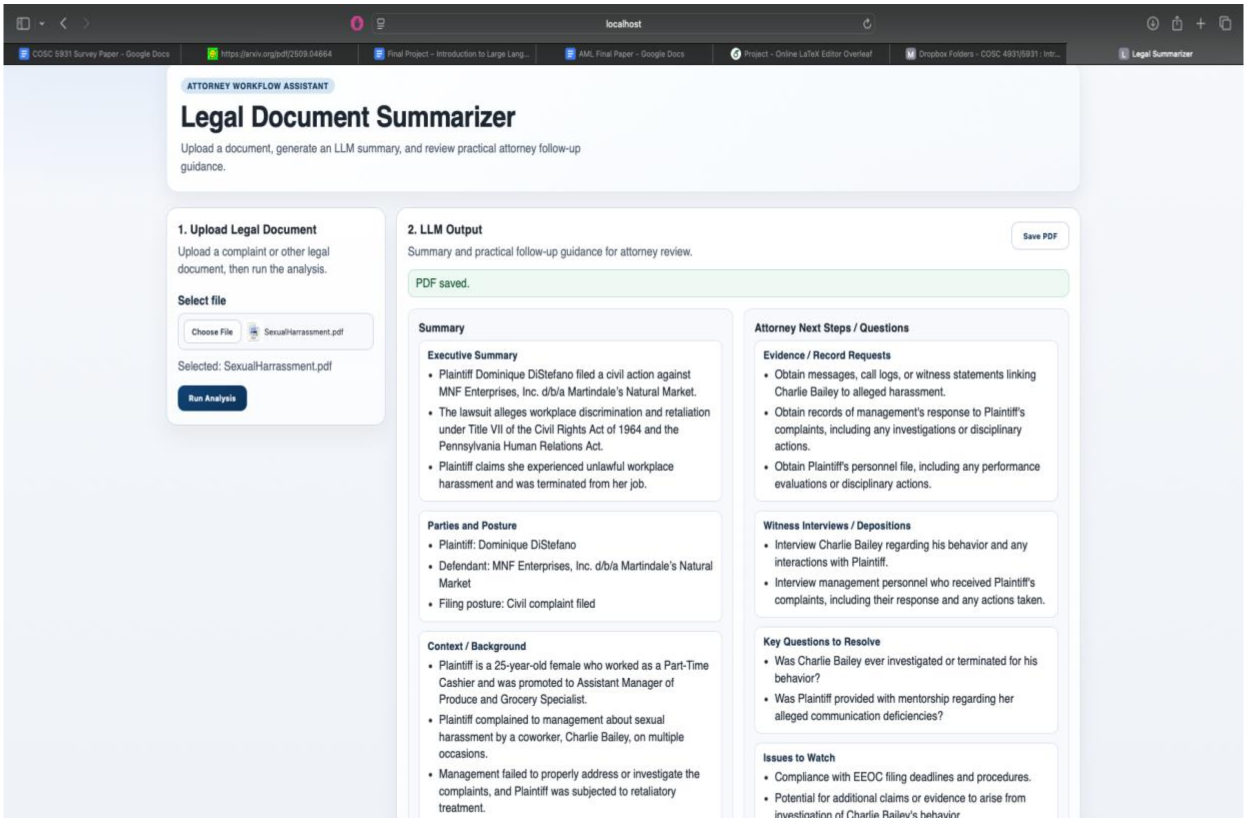
The LLM process uses prompt chaining. The first prompt generates the case summary from the extracted document text. It requires fixed Markdown headings: **Executive Summary**, **Parties and Posture**, **Context / Background**, **Main Claims / Charges**, **Deadlines and Time-Sensitive Items**, and **Missing Facts**. These headings were chosen because they match what a legal reviewer needs during early case triage: a quick overview, party identification, procedural status, claims, time-sensitive items, and gaps in the document.

The second prompt generates attorney next steps from the summary rather than directly from the original complaint. This prompt asks for practical follow-up categories such as **Evidence / Record Requests**, **Witness Interviews / Depositions**, **Key Questions to Resolve**, and **Issues to Watch**. Splitting the task into two prompts helped to control the output. The first prompt focuses on understanding the document, while the second prompt focuses on what a legal professional might do next based on that understanding.

Prompt engineering was one of the most important methods in the project, and the prompts were designed iteratively in collaboration with a human attorney. This helped make the instructions more closely match what a legal professional would actually need during early case review. The prompts use role prompting by telling the model it is assisting counsel who has received the document for review. They also use structured output prompting by requiring exact headings, short bullet points, no extra sections, and no long paragraphs, all designed for easier reading. The prompts also instruct the model to avoid inventing facts, avoid repeating the same fact across sections, preserve names and citations, and use “Unclear from provided text” only when the document truly does not answer a point.

Additionally, the prompts mainly use zero-shot prompting. This means that the model receives detailed instructions and the document text but not a full example answer. This choice kept the prompt flexible across case types, but it also made the wording of the instructions very important. Several prompt constraints were iteratively added to address frequently observed problems during testing, such as the model asking questions that were already answered in the complaint, treating background facts as legal claims, or giving generic litigation advice instead of document-specific next steps.

Figure 1: LLM Legal Complaint Analyzer Example



Dataset

The evaluation dataset consisted of 10 publicly available federal legal complaints sourced from PACER, an electronic database service that provides nationwide access to U.S. federal legal documents and case information. The dataset was carefully curated to ensure that the tool would be tested across multiple legal contexts rather than only one narrow type of complaint or type of law. This variety was important because a legal document summarizer should be able to handle different structures, factual patterns, and legal vocabularies.

The dataset included employment and workplace disputes, civil rights complaints, government-related matters, criminal complaints, and business or intellectual property disputes (see Table 1 for a full list).

Table 1: Dataset Case Types

Case Type	Case Summaries
Employment / Workplace Disputes	• Workplace sexual harassment + retaliation • Unpaid commissions + breach of contract
Civil Rights	• Americans with Disabilities Act (ADA) violation • Discrimination based on national origin
Government	• Immigration / ICE • Whistleblowing / Treason*
Criminal Law	• Drug trafficking + firearms • Human trafficking + obstruction of justice
Business / Intellectual Property	• Trademark infringement • Class action*

* = Excluded from final evaluation

The dataset included both relatively straightforward complaints and more complex filings. Some documents involved a narrow legal theory with a clearer factual timeline, while others included multiple parties, many factual allegations, several statutes or counts, and more specialized legal issues. This helped reveal where the tool performed well and where it struggled. For example, a narrower Freedom of Information Act (FOIA) or employment complaint was easier for the model to summarize than an intellectual property dispute with many products and counts or a criminal complaint where the most important facts were related to evidence, warrants, and charges.

Each case folder (evals/cases) in the GitHub repository contains the original case document, the generated output from the tool, and attorney review or rating worksheets used for human evaluation purposes. Two cases, a class action case and a whistleblowing-related case, were included in the larger collection of cases but were ultimately excluded from evaluation because the tool could not process them successfully. These documents were scanned filings, meaning they functioned more like images than text-based PDFs. Since the current version of the tool does not include image processing or OCR capabilities, it could not reliably extract the text needed to generate usable outputs.

The dataset was not intended to be large enough for statistical proof of model quality. Instead, it functioned as a practical evaluation set meant to evaluate what an LLM can do without excessive fine-tuning or training. It also allowed the tool to be tested against realistic documents and made it possible to compare the model output against attorney-reviewed expectations.

Evaluation

The evaluation process combined tool output with qualitative human review. A human attorney (a litigator specializing in employment law) reviewed the complaints and filled out worksheets describing the important case information, including parties, posture, claims, background facts, and review notes. The same complaints were run through the tool, and the generated outputs were saved in Markdown files for comparison. The attorney then evaluated the outputs using five categories on a 1 to 5 scale.

The five evaluation categories were **Factual Accuracy**, **Completeness**, **Legal Precision**, **Attorney Usefulness**, and **Actionability**. **Factual Accuracy** measured whether the output correctly reflected what the complaint actually said. **Completeness** measured whether the output captured the most important facts, claims, parties, dates, and legal issues. **Legal Precision** measured whether the output described claims, posture, charges, and legal framing correctly. **Attorney Usefulness** measured whether the output would help an attorney quickly understand the case. **Actionability** measured whether the next steps and questions were concrete, practical, and tied to the document.

The results showed that the tool was strongest as a first-pass summarizer. It usually identified the broad case type, parties, posture, and main claims or charges correctly. This suggests that LLMs can excel at quickly orienting a reviewer who has not yet read the full complaint. However, the tool was less reliable when the task required deciding which facts were most attorney-critical or generating next steps tailored to a particular legal role.

Table 2: Evaluation Results

Category	Average Score	Main Takeaway
Factual Accuracy	4.25 / 5	Strongest category; the tool usually reflected the complaint correctly.
Completeness	3.25 / 5	Mixed performance; important facts, dates, products, evidence, or claim details were sometimes missed.
Legal Precision	3.5 / 5	Generally adequate, but the tool sometimes misstated posture, charges, statutory bases, or claim timing.
Attorney Usefulness	3 / 5	Helpful for initial orientation, but not consistently enough for attorney-ready litigation planning.
Actionability	2.88 / 5	Weakest category; next steps were sometimes generic, misdirected, or already answered by the complaint.

The overall average across the five categories was 3.38 out of 5. The highest-rated area was factual accuracy, which is encouraging because factual reliability is the baseline requirement for any legal tool. The weakest area was actionability, which shows that generating useful legal next steps is more difficult than summarizing the document.

Case-level results showed clear patterns. The tool performed best on shorter or cleaner complaints with a narrow legal theory, such as the immigration and sexual harassment cases. It was generally able to correctly identify parties, posture, legal category, and the main dispute. In more complex cases, the output often became too generic. The ADA violation case output missed specific access barriers and facility details. The drug trafficking case output identified the basic charges, but missed important criminal defense issues such as the search warrant angle, drug quantity, firearm details, and suppression-relevant facts. The trademark infringement case output was directionally correct but did not name enough accused products, marks, patents, or counts. The human trafficking case output captured the overall context but missed central kidnapping facts and evidence.

The evaluation also showed that a summary can be mostly accurate but still not fully useful for legal work. Attorneys often need the specific facts that shape strategy, not just the general type of case. For example, saying that a complaint involves discrimination is less useful than identifying the specific incidents, dates, supervisors, policies, and protected-class allegations. Similarly, in a criminal case, the next steps should focus on discovery, warrants, probable cause, chain of custody, and evidence issues rather than generic interview suggestions.

Another recurring issue was audience confusion. The tool sometimes did not clearly understand whether the output was intended for plaintiff's counsel, defense counsel, prosecution, criminal defense, or a neutral reviewer. This affected the next-steps section because each legal role has different priorities. A plaintiff-side employment attorney, a defense attorney, and a criminal defense attorney would not ask the same first questions or request the same documents.

Discussion of Limitations + Future Improvements

The main limitation of the current tool is document extraction. The system works best when the uploaded file already contains extractable text, such as a text-based PDF, DOCX file, or TXT file. Scanned documents were much more difficult for the tool to process because they are image-based, and the tool does not currently include OCR or image-processing capabilities. This mattered during evaluation because some scanned filings could not be processed successfully, and two cases had to be excluded from evaluation because of this. Since many real legal filings, exhibits, and older court documents are scanned, adding OCR would be one of the most important future improvements.

Another limitation is the size of the evaluation dataset. Ten federal legal complaints were enough to test the prototype across several case types, but they are not enough to prove broad reliability. Legal documents vary by court, jurisdiction, practice area, formatting, and document type. Future testing should include more complaints, motions, orders, contracts, administrative filings, state-court documents, and scanned exhibits. A larger dataset would make it easier to tell whether the tool's weaknesses are consistent patterns or mostly tied to a few difficult cases.

The evaluation also showed that the tool's output quality could be improved in two related areas: detail extraction and role awareness. The model often identified the correct legal category, but it sometimes missed details that would make the summary more attorney-ready, such as specific dates, parties, statutes, counts, product names, evidence, requested relief, or deadlines. It also sometimes asked questions that were already answered or implied somewhere else in the complaint. Future prompt engineering should continue focusing on concrete fact extraction and stronger grounding in the source document. A future version of the application could also include an attorney role selector, such as plaintiff's counsel, defense counsel, prosecutor, criminal defense, neutral evaluator, or unknown, so that next-step recommendations are better tailored to each user's perspective.

Additional future improvements could make the tool more useful as a legal workflow product. LLaMA 3.1 8B could be compared against larger models or models better suited for legal and long-document tasks. The interface could also add features such as allowing users to edit or annotate generated output, highlighting the source text that supports each summary bullet, and exporting a more polished attorney-review memo. Source highlighting would be especially useful because it would help attorneys verify the model's output rather than relying on the summary alone.

Conclusion

Overall, this project shows that an LLM can be very useful as a legal triage assistant, especially for a quick first-pass review of a complaint. The tool was strongest at identifying the broad shape of a case, including the parties, procedural posture, case type, and main claims or charges. This suggests that LLMs can help reduce some of the time spent on initial document review, as long as the output is treated as a starting point rather than a final legal analysis.

At the same time, the evaluation showed that attorney-ready work requires more than a generally accurate summary. The tool still needs stronger detail extraction, better grounding in the source document, role-specific next-step recommendations, and improved support for scanned filings. These limitations are important, but they also point clearly toward future improvements. With OCR, expanded testing, stronger prompt design, and more workflow-focused features, the tool could become a more practical assistant for early case assessment.